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Cryptography ,a tool to protect your data.

Abstract:

Cryptography is the process of encrypting messages and other data in order to transmit them in a form that can only be accessed by the intended recipients. It was initially applied to written messages. With the introduction of modern computers, cryptography became an important tool for securing many types of digital data.

PRINICIPAL TERMS

- **hash function:** an algorithm that converts a string of characters into a different, usually smaller, fixed-length string of characters that is ideally impossible either to invert or to replicate.
- **public-key cryptography:** a system of encryption that uses two keys, one public and one private, to encrypt and decrypt data.
- **substitution cipher:** a cipher that encodes a message by substituting one character for another.
- **symmetric-key cryptography:** a system of encryption that uses the same private key to encrypt and decrypt data.
- **transposition cipher:** a cipher that encodes a message by changing the order of the characters within the message.

WHAT IS CRYPTOGRAPHY?

The word "cryptography" comes from the Greek words *kryptos* ("hidden," "secret") and *graphein* ("writing"). Early cryptography focused on ensuring that written messages could be sent to their intended recipients without being intercepted and read by other parties. This was achieved through various encryption techniques. Encryption is based on a simple principle: the message is transformed in such a way that it becomes unreadable. The encrypted message is then transmitted to the recipient, who reads it by transforming (decrypting) it back into its original form.

Introduction:

Cryptography is the science of using mathematics to encrypt and decrypt data. Cryptography enables you to store sensitive information or transmit it across insecure networks (like the Internet) so that it cannot be read by anyone except the intended recipient. While cryptography is the science of securing

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data, cryptanalysis is the science of analyzing and breaking secure communication. Classical cryptanalysis involves an interesting combination of analytical reasoning, application of mathematical tools, pattern finding, patience, determination, and luck. Cryptanalysts are also called attackers. Cryptology embraces both cryptography and cryptanalysis

Techniques:

There are several techniques for implementing cryptography algorithm. For purposes of this paper, they will be categorized based on the number of keys that are employed for encryption and decryption, and further defined by their application and use. The three types of algorithms that will be discussed are (Figure 1):

- *Secret Key Cryptography (SKC):* Uses a single key for both encryption and decryption; also called *symmetric encryption*. Primarily used for privacy and confidentiality.
- *Public Key Cryptography (PKC):* Uses one key for encryption and another for decryption; also called *asymmetric encryption*. Primarily used for authentication, non-repudiation, and key exchange.
- *Hash Functions:* Uses a mathematical transformation to irreversibly "encrypt" information, providing a digital fingerprint. Primarily used for message integrity.

Out of all these cryptographic algorithms, some of the these algorithms are widely used for encryption like caesar cipher, monoalphabetic cipher ,polyalphabetic cipher, playfair cipher, hill cipher etc. Here we will discuss some of these cryptographic algorithms briefly.

The oldest and the most common cryptographic algorithm used for encryption is caesar cipher. First,we are going to discuss about that.

Caesar cipher:

The Caesar cipher is a classic example of ancient cryptography and is said to have been used by Julius Caesar. The Caesar cipher is based on transposition and involves shifting each letter of the plain text by a certain number of letters, historically three. The cipher text can be decrypted by applying the same number of shifts in the opposite

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direction. This type of encryption is known as a substitution cipher, due to the substitution of one letter for another in a consistent fashion.

Monoalphabetic cipher:

Monoalphabetic cipher is a substitution cipher in which for a given key, the cipher alphabet for each plain alphabet is fixed throughout the encryption process. For example, if 'A' is encrypted as 'D', for any number of occurrence in that plaintext, 'A' will always get encrypted to 'D'.

Playfair cipher:

The Playfair cipher code is a technique of symmetry encryption. The simple description of the Playfair cipher would be it is a written code and it became the first technique was used for the encryption of data. It was first introduced in the year 1854. It has all the keys which are arranged in alphabetical letters or geometric patterns so that the decryption of Playfair cipher can be done .

The Playfair cipher is also referred to as Playfair square. It was created by Sir Charles Wheatstone. But it was named after a lot of affairs he was the one who promoted the uses of it. It was then used by the Britishers and the Australians in the world war and to retain privacy on certain messages which layer to be transferred from one place to another.

Discussion:

Caesar cipher is simple for vulnerable to attack because of it's less number of possible keys. Monoalphabetic cipher is better than Caesar cipher in terms of attacks but still then it suffers from cryptanalysis because of language characteristic if the language is known to cryptanalysis he can decrypt the encrypted text by matching with the frequency of the repeated letter. One approach to improve security against attack is to encrypt multiple letters in a plain text at one time. Playfair cipher uses this approach and hence it considered to be better than above two techniques. It treats diagrams in plaintext as a single unit and translates these units in to cipher text diagrams.

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Pros and cons:

Benefits of cryptography:

Cryptography is an essential information security tool. It provides the four most basic services of information security −

- **Confidentiality** − Encryption technique can guard the information and communication from unauthorized revelation and access of information.
- **Authentication** − The cryptographic techniques such as MAC and digital signatures can protect information against spoofing and forgeries.
- **Data Integrity** − The cryptographic hash functions are playing vital role in assuring the users about the data integrity.
- **Non-repudiation** − The digital signature provides the non-repudiation service to guard against the dispute that may arise due to denial of passing message by the sender.

Drawbacks of cryptography:

Apart from the four fundamental elements of information security, there are other issues that affect the effective use of information −

- A strongly encrypted, authentic, and digitally signed information can be **difficult to access even for a legitimate user** at a crucial time of decision-making. The network or the computer system can be attacked and rendered non-functional by an intruder.
- **High availability,** one of the fundamental aspects of information security, cannot be ensured through the use of cryptography. Other methods are needed to guard against the threats such as denial of service or complete breakdown of information system.
- Another fundamental need of information security of **selective access control** also cannot be realized through the use of cryptography. Administrative controls and procedures are required to be exercised for the same.

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Conclusion and Future Scope:

Elliptic Curve Cryptography (ECC) has already been invented but its advantages and disadvantages are not yet fully understood. ECC allows to perform encryption and decryption in a drastically lesser time, thus allowing a higher amount of data to be passed with equal security. However, as other methods of encryption, ECC must also be tested and proven secure before it is accepted for governmental, commercial, and private use.